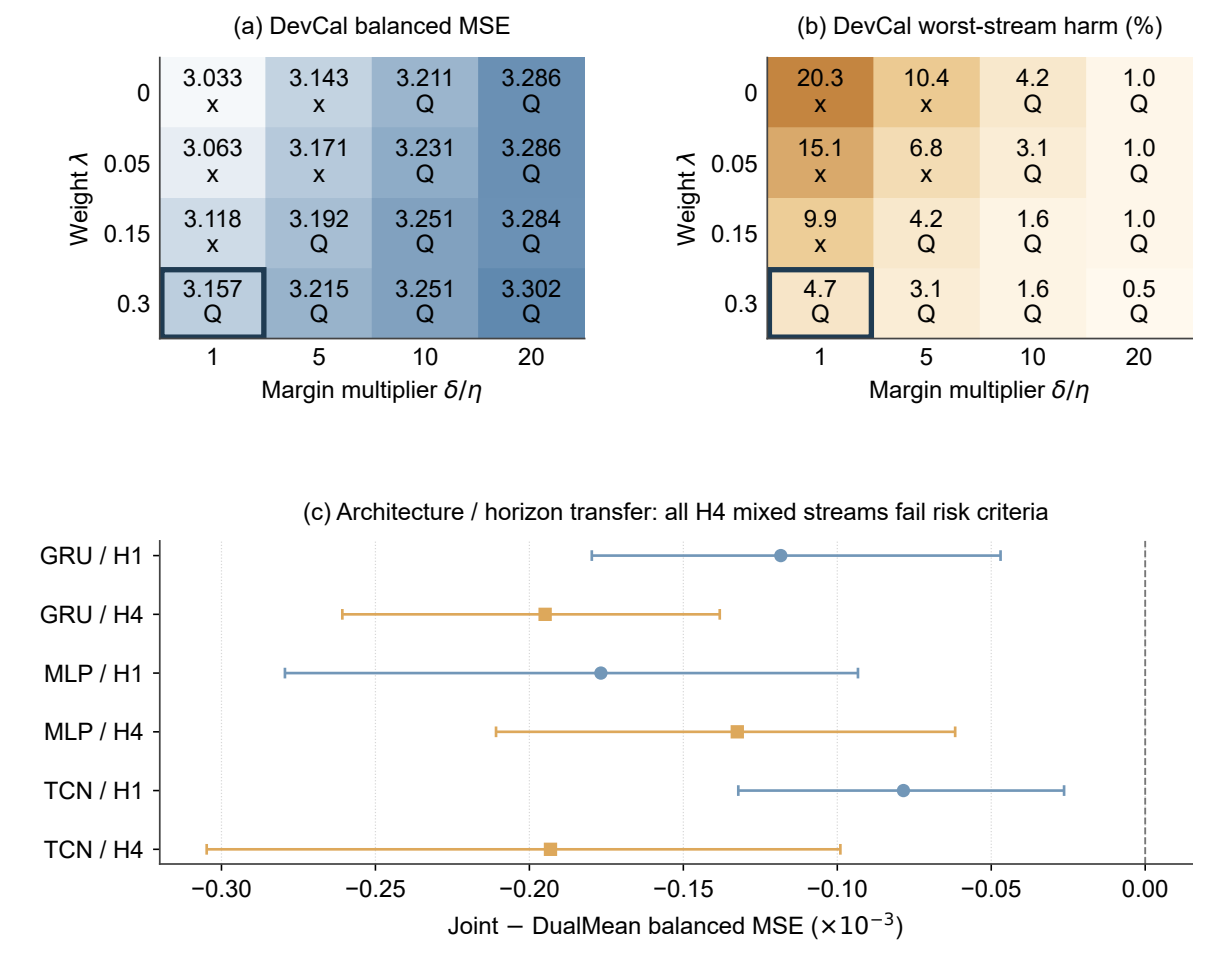
(a) DevCal balanced MSE
Weight λ
0
0.05
0.15
0.3
3.033 x
3.143 x
3.211 Q
3.286 Q
3.063 x
3.171 x
3.231 Q
3.286 Q
3.118 x
3.192 Q
3.251 Q
3.284 Q
3.157 Q
3.215 Q
3.251 Q
3.302 Q
1
5
10
20
Margin multiplier δ/η
(b) DevCal worst-stream harm (%)
Weight λ
0
0.05
0.15
0.3
20.3 x
10.4 x
4.2 Q
1.0 Q
15.1 x
6.8 x
3.1 Q
1.0 Q
9.9 x
4.2 Q
1.6 Q
1.0 Q
4.7 Q
3.1 Q
1.6 Q
0.5 Q
1
5
10
20
Margin multiplier δ/η
(c) Architecture / horizon transfer: all H4 mixed streams fail risk criteria
GRU / H1
GRU / H4
MLP / H1
MLP / H4
TCN / H1
TCN / H4
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Joint − DualMean balanced MSE (×10⁻³)